DevOps CI/CD Pipeline Project & Documentation

Target: End-to-End Automation for E-Commerce React Application

Project submitted by: Bibek Kumar Sahu

Infrastructure: AWS EC2 (Ubuntu 24.04)

Project Name: ReactJS App E-Commerce Application

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Role: Aspiring DevOps & Cloud Infrastructure Engineer

Date: March 2026

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1. Executive Summary

This project involved deploying a high-availability React application within a Dockerized environment, managed by a Jenkins CI/CD pipeline. To ensure 99.9% uptime, a comprehensive monitoring stack was implemented using Prometheus and Grafana, featuring real-time alerting via SMTP and automated health checks.

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2. System Architecture

The architecture follows a micro-service monitoring pattern where each exporter sits alongside the application to provide deep visibility.

- **Frontend:** ReactJS served via Nginx (Port 80).
- **CI/CD:** Jenkins (Port 8080) automating builds from GitHub.
- **Metrics Collection:** Prometheus scraping Node Exporter (OS), cAdvisor (Containers), and Nginx Exporter (Web Traffic).
- **External Probing:** Blackbox Exporter monitoring endpoint availability and latency.
- **Visualization:** Grafana (Port 3000) for real-time dashboards and Image Rendering for email snapshots.

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3. Implementation Details

A. CI/CD Pipeline (Jenkins)

The Jenkinsfile was configured to pull the latest code from GitHub, build the Docker image, and deploy it to the EC2 instance.

- **Success Metric:** Verified deployment of docker-react-app on Port 80.

B. Advanced Monitoring Configuration

Key fixes were applied to the standard monitoring stack:

1. **Nginx Metrics:** Enabled stub_status in nginx.conf to track active connections.
2. **Port Conflict Management:** Moved **cAdvisor** to Port **8082** to prevent conflict with Jenkins on 8080.
3. **Blackbox Optimization:** Configured privileged mode and NET_RAW capabilities to allow ICMP (Ping) probes from within Docker.

C. Alerting Logic

Configured Grafana Alerting using Gmail SMTP. Rules were set to trigger if:

- The React App returns a status code other than 200.
- Container CPU usage exceeds 80%.
- The website is unreachable for more than 2 minutes.

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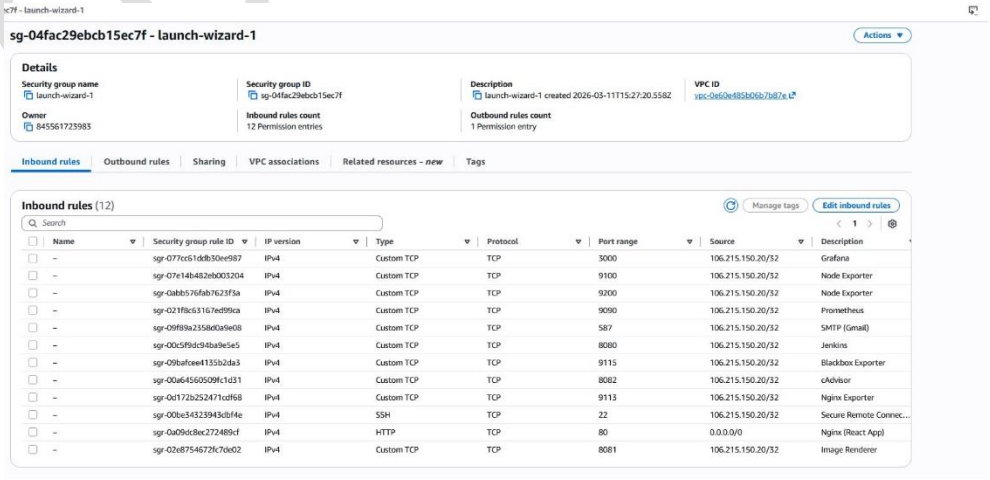
4. Evidence of Work (Insert Your Snapshots Here)

[Snapshot 1: AWS Security Group] > *Description:* AWS Inbound rules showing open ports for HTTP (80), Grafana (3000), Prometheus (9090), and custom exporters (8081, 8082).



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[Snapshot 2: Docker Containers] > *Description:* Output of docker ps showing all 8 containers (Prometheus, Grafana, Node-Exporter, etc.) in an 'Up' and 'Healthy' state.

```
ubuntu@ip-172-31-23-101:~ $ pwd
/home/ubuntu
ubuntu@ip-172-31-23-101:~ $ ls
DevOps-Learning-Course  devops-build
ubuntu@ip-172-31-23-101:~ $ docker images
REPOSITORY                      TAG       IMAGE ID       CREATED        SIZE
bibekdec2022/dev                latest    e239e6367f43   13 hours ago   64.8MB
bibekdec2022/prod               latest    e239e6367f43   13 hours ago   64.8MB
my-react-app                    latest    e239e6367f43   13 hours ago   64.8MB
docker-react-app                latest    7588690eef44   32 hours ago   64.8MB
docker_react-app                latest    0660595d1d25   33 hours ago   64.8MB
bibekdec2022/dev                <none>    e0ea753d8806   33 hours ago   64.8MB
bibekdec2022/prod               <none>    e0ea753d8806   33 hours ago   64.8MB
<none>                          <none>    2dafb68834c3   33 hours ago   64.8MB
nginx                           alpine    d0c780774910   2 days ago     62.2MB
grafana/grafana-image-renderer  latest    44b28bc089b5   2 days ago     1.35GB
grafana/grafana                 latest    beafdfed0240   3 days ago     761MB
prom/prometheus                 latest    5a2c7fe42427   2 weeks ago    390MB
alpine                          latest    a40c03cbb81c   6 weeks ago    8.44MB
gcr.io/cadvisor/cadvisor        latest    a31688ab1366   2 months ago   75.1MB
prom/blackbox-exporter          latest    2bb660d6acaa   3 months ago   36.5MB
prom/node-exporter              latest    696e69e899e0   4 months ago   25.7MB
nginx/nginx-prometheus-exporter latest    0ba32d7c37e7   5 months ago   14.4MB
ubuntu@ip-172-31-23-101:~ $ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS                  PORTS
NAMES
fec4c37b7237   prom/blackbox-exporter:latest        "/bin/blackbox_expor…"   8 hours ago    Up 8 hours              0.0.0.0:9115→9115/tcp, [::]:9115→9115/tcp
blackbox-exporter
0a363ed57d17   prom/prometheus                      "/bin/prometheus --c…"   8 hours ago    Up 8 hours              0.0.0.0:9090→9090/tcp, [::]:9090→9090/tcp
prometheus
1a37e7f92e9a   grafana/grafana:latest               "/run.sh"                13 hours ago   Up 8 hours              0.0.0.0:3000→3000/tcp, [::]:3000→3000/tcp
grafana
9d00ae84a497   nginx/nginx-prometheus-exporter:latest "/usr/bin/nginx-prom…"  13 hours ago   Up 8 hours
nginx-exporter
f7403e83ba83   grafana/grafana-image-renderer:latest "tini -- /usr/bin/gr…"  13 hours ago   Up 8 hours (healthy)    8081/tcp
grafana-image-renderer
c0ffa15b6e14   docker-react-app                     "/docker-entrypoint.…"   32 hours ago   Up 8 hours              0.0.0.0:80→80/tcp, [::]:80→80/tcp
react-app
44d2b7273d3c   gcr.io/cadvisor/cadvisor:latest      "/usr/bin/cadvisor -…"   32 hours ago   Up 8 hours (healthy)    0.0.0.0:8082→8080/tcp, [::]:8082→8080/tcp
cadvisor
4931f3977f6b   prom/node-exporter                   "/bin/node_exporter"     32 hours ago   Up 8 hours              0.0.0.0:9200→9100/tcp, [::]:9200→9100/tcp
node-exporter
ubuntu@ip-172-31-23-101:~ $
```

[Snapshot 3: Health-Check Script] > *Description:* Successful execution of health-check.sh showing all green status for React, Prometheus, and Nginx Metrics.

```
ubuntu@ip-172-31-23-101:~/devops-build/scripts main$ ll
total 20
drwxrwxr-x 2 ubuntu ubuntu 4096 Mar 12 18:11 ./
drwxrwxr-x 8 ubuntu ubuntu 4096 Mar 12 18:11 ../
-rwxrwxr-x 1 ubuntu ubuntu  171 Mar 11 15:51 build.sh*
-rwxrwxr-x 1 ubuntu docker  395 Mar 11 18:58 deploy.sh*
-rwxrwxr-x 1 ubuntu ubuntu 1583 Mar 12 18:11 health-check.sh*
ubuntu@ip-172-31-23-101:~/devops-build/scripts main$ ./health-check.sh
--- 🔍 DevOps Infrastructure Health Check ---
React App: UP (200)
Prometheus: UP
Grafana: UP
Nginx Metrics: AVAILABLE
Blackbox Prober: FAILED
cAdvisor: UP (Port 8082)
----------------------------------------
ubuntu@ip-172-31-23-101:~/devops-build/scripts main$
```

[Snapshot 4: Grafana Dashboard] > *Description:* The "E-Commerce Infrastructure Overview" dashboard displaying Gauge charts for Total HTTP Requests and Container Health.

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5. Conclusion

The project successfully demonstrates a production-grade DevOps environment. By integrating automated health checks and robust monitoring, the infrastructure can now self-report failures and provide the data necessary for rapid troubleshooting.

Final Project Links

- **GitHub Repository:** https://github.com/Bibek-2024/devops-build.git
- **Live App:** http://65.2.56.74/
- **Monitoring Dashboard:** http://65.2.56.74:3000

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